Physics 4B Spring 2026
Week 5 Discussion

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1. **Brainwarmer**[1]. Find the lift on an airplane wing of area 50 m^2 if the velocity of air at the upper and lower parts is 90 m/s and 70 m/s, respectively. Assume $\rho_{\text{air}} = 1.16\ \text{kg/m}^3$.

   Solution: We start with Bernoulli's law

   $$P_{\text{u}} + \frac{1}{2}\rho_{\text{u}} v_{\text{u}}^2 + \rho_{\text{u}} g h_{\text{u}} = P_{\text{l}} + \frac{1}{2}\rho_{\text{l}} v_{\text{l}}^2 + \rho_{\text{l}} g h_{\text{l}} \tag{1}$$

   We can assume the wing is thin enough so that $h_1 \approx h_2$, and we can neglect the potential terms. We can additionally assume the air density is the same on both sides so $\rho_{\text{u}} = \rho_{\text{l}} = \rho_{\text{air}}$. Then we have

   $$\Delta P = P_{\text{l}} - P_{\text{u}} = \frac{1}{2}\rho_{\text{air}}(v_{\text{u}}^2 - v_{\text{l}}^2) \tag{2}$$

   The lift on the plane is the force due to this pressure difference, so

   $$F = \Delta P \cdot A = \frac{1}{2}\rho_{\text{air}} A(v_{\text{u}}^2 - v_{\text{l}}^2) \tag{3}$$

2. **Atmosphere density**[2]. Assume the atmosphere at height h above sea level has pressure $P(h)$ and density $\rho(h)$ and that the ratio $P(h)/\rho(h)$ is independent of h (that would happen if the temperature doesn't change with h). Assume at sea level $P = P_0 = 101,325$ Pa and $\rho = \rho_0 = 1.204\ \text{kg/m}^3$.

   (a) What would be the height of the atmosphere if its density was constant, equal to ρ_0?

   (b) Derive equations for $\rho(h)$ and $P(h)$ assuming g is independent of h. Write your answers in terms of the height found in (a), call it H.

   (c) Same as (b) taking into account the variation of g with h. You may assume $h \ll R$, with R the earth radius.

   (d) Compare the results for the density at a given height h assuming g constant versus g decreasing with h. Does the density decrease, increase or stay the same?

[1] Shankar 10.9
[2] From additional homework problem

(e) There are 2 different ways to interpret and answer (d). One is to compare the two situations for the same ρ_0, the other to compare them for the same total number of air molecules. Discuss the two scenarios.

Solution:

(a) Let the height of the atmosphere be H. If we held the density constant, then $P(h) = P_0 - \rho_0 g h$. At the top of the atmosphere (i.e. $h = H$), the absolute pressure is 0, so $P_0 = \rho_0 g H$, which leads to $H = \frac{P_0}{\rho_0 g}$.

(b) We knew in the last part that the pressure P was linear in h because the density was constant. We don't know the exact dependence of $P(h)$ on h this time, but we still have the relationship $\Delta P = -\rho g \Delta h$ (which comes from our derivation of the hydraulic pressure). Taking the limit of this, we get the hydrostatic equilibrium equation

$$\frac{dP}{dh} = -\rho g \tag{4}$$

Since we assume $P(h)/\rho(h)$ stays constant, we can replace $\rho(h)$ with $\rho_0 P(h)/P_0 = P(h)/H$. The equation then becomes

$$\frac{dP}{dh} = -\frac{P}{H} \tag{5}$$

which has solution $P(h) = P_0 e^{-h/H}$. The density would be $\rho(h) = \rho_0 e^{-h/H}$. Note this result is based on the assumption that the temperature stays constant at different heights.

(c) We have $g \propto 1/R^2$, so $g(h) = g_0(R/(R+h))^2 \approx g_0 R^2(1 - 2h/R)$. The hydrostatic equilibrium equation in the previous part becomes

$$\frac{dP}{dh} = -\frac{P}{H}\left(1 - \frac{2h}{R}\right) \tag{6}$$

where we redefined $H = P_0/\rho_0 g$.

This has solution

$$P(h) = P_0 e^{-\frac{h}{H} + \frac{h^2}{HR}} \tag{7}$$

so the density is $\rho(h) = \rho_0 e^{-\frac{h}{H} + \frac{h^2}{HR}}$. This decreases more slowly with height than the previous result, which makes sense because gravity becomes weaker as the height increases, so the air is less compressed to the bottom.

(d) As we concluded from last part, the density at height h increases if we allow g to vary with height.

(e) We have already discussed the case with fixed ρ_0. To keep the total number of air particles the same, we need to fix the total mass, i.e.

$$\int_0^\infty dh\ \rho(h) \tag{8}$$

is the same for both cases. For the constant g case,

$$\int_0^\infty dh\ \rho_{\text{const g}}(h) = \rho_0 H \tag{9}$$

For the variable g case,

$$\int_0^\infty dh\ \rho_{\text{var g}}(h) > \rho_0 H \tag{10}$$

To keep the two equal, we must have $\rho_{0,\text{var g}} < \rho_{0,\text{const g}}$.

3. **Buoyancy**[3]. A body of density ρ is dropped from a height H into a lake where the water density is $\rho_w > \rho$.

   (a) To what depth d will it sink before beginning to rise up?

   (b) When it floats, what fraction f of its volume will be submerged?

   Solution:

   (a) We first need to figure out at speed the object enters the water. By kinematics, we have $v^2 = 2gH$. Once it enters the water, we again need to use kinematics to figure out the distance it travels. Since the problem didn't give us the dimensions of the object, we can assume that it wants us to ignore the part where the object is partially in the water (and therefore experiences less-than-maximum buoyancy). Therefore, we will just consider when the object is fully submerged, and use the buoyancy in this case to calculate the acceleration of the object.

   When fully submerged, the net force on the object is

$$F_{\text{net}} = mg - \rho_w g V = (\rho - \rho_w) g V \tag{11}$$

   so the acceleration on the object is

$$a = \frac{F_{\text{net}}}{m} = \frac{\rho - \rho_w}{\rho} g \tag{12}$$

[3] Shankar 10.5

which is a constant. We then use the same kinematic equation to find the maximum depth.

$$d = \frac{v^2}{2|a|} = \frac{2gH}{2g}\frac{\rho}{\rho_w - \rho} = \frac{\rho H}{\rho_w - \rho} \tag{13}$$

I took the magnitude of a here because it's negative (i.e. downward).

Alternative solution: We made the simplifying assumption that the object's thickness can be ignored. What if we consider it? Let's assume the object is rectangular and has thickness h. We can either repeat the previous steps (but this time there are three stages of motion instead of two), or use conservation of energy to make the calculation easier. We'll do the latter. I will consider all heights to be the heights of the bottom of the object.

Conservation of energy says that

$$\frac{1}{2}mv_1^2 + mgH + W = \frac{1}{2}mv_2^2 - mgd \tag{14}$$

Here W is the work done by non-gravity forces on the object, and the kinetic energy terms are equal to 0 because the initial final states have zero velocity. The non-gravity forces as a function of the height y above the surface is (we'll define positive direction to be pointing up)

$$F_{\text{ng}}(y) = \begin{cases} 0 & 0 < y < H \\ -\rho_w V g \frac{y}{h} & -h < y < 0 \\ \rho_w V g & -d < y < -h \end{cases} \tag{15}$$

The work done by this force on the object is

$$\begin{aligned} \int_H^{-d} \vec{F}_{\text{ng}} \cdot d\vec{y} &= \int_H^{-d} F_{\text{ng}}(y)dy = -\frac{\rho_w V g}{h}\int_0^{-h} dy\ y + \rho_w V g\,(-d+h) \\ &= -\frac{1}{2}\rho_w V g h + \rho_w V g\,(-d+h) \\ &= -\rho_w V g\left(d + \frac{h}{2}\right) = -mg\frac{\rho_w}{\rho}\left(d + \frac{h}{2}\right) \end{aligned} \tag{16}$$

so we end up with

$$mgH - mg\frac{\rho_w}{\rho}\left(d + \frac{h}{2}\right) = -mgd \tag{17}$$

and the final answer is

$$d = \frac{\rho H + \frac{1}{2}\rho_w h}{\rho_w - \rho} \tag{18}$$

This reduces to our previous result answer if you take $h \to 0$. Additionally, if you take all heights to be the heights of the center of mass of the object, you will get exactly our previous answer.

(b) When it's floating, the buoyancy balances the weight, so we have

$$\rho_w g V_{\text{submerged}} = \rho V g \tag{19}$$

The fraction of submerged volume is

$$f = \frac{V_{\text{submerged}}}{V} = \frac{\rho}{\rho_w} \tag{20}$$

4. **Bernoulli**[4]. Consider a U-tube with both limbs open to the atmosphere with equal heights of mercury $\rho_{\text{Hg}} = 13.6 \times 10^3$ kg/m^3. When the air ($\rho_a = 1$ kg/m^3) at speed v begins to flow above just one limb, the mercury there rises by 4 cm above the old level. What is v? (Remember, one limb's gain is the other limb's loss.)

Solution: We start with Bernoulli's law

$$P_1 + \frac{1}{2}\rho_1 v_1^2 + \rho_1 g h_1 = P_2 + \frac{1}{2}\rho_2 v_2^2 + \rho_2 g h_2 \tag{21}$$

This problem is tricky because it involves applying Bernoulli's law at two different places: (i) At the openings of the tube on both ends, and (ii) at the liquid surfaces on both ends.

(i) At the openings (equal heights), we have

$$P_1 + \frac{1}{2}\rho_a v_1^2 = P_2 + \frac{1}{2}\rho_a v_2^2 \tag{22}$$

with $v_2 = 0$ and $v_1 = v$. I have replaced ρ_1 and ρ_2 with the density of air since we are looking at the tube openings. This creates a pressure difference

$$\Delta P = P_2 - P_1 = \frac{1}{2}\rho_a v^2 \tag{23}$$

(ii) At the liquid surfaces: If we assume the tube is symmetrical in shape, then a rise of 4 cm on one side means a drop of 4 cm on the other side. This creates a height difference of $\Delta h = h_{\text{u}} - h_{\text{l}} = 8$ cm. There are no fluid flows because the mercury isn't flowing in a persistent way (it settles once the equilibrium is reached), but there is a pressure difference due to what we just considered in (i).

[4]Shankar 10.15

Removing the kinetic terms, Bernoulli's law then says

$$P_1 + \rho_{\mathrm{Hg}} g h_1 = P_2 + \rho_{\mathrm{Hg}} g h_2 \tag{24}$$

Note that here the density is that of mercury, not air, because the pressure in the liquid has nothing to do with air! We can then equate the pressure differences - the ΔP due to the flowing air at the opening is the cause of the change in the mercury levels:

$$\Delta P = \rho_{\mathrm{Hg}} g \Delta h = \frac{1}{2} \rho_a v^2 \tag{25}$$

Putting everything together, we get

$$v = \sqrt{\frac{2 \rho_{\mathrm{Hg}} g \Delta h}{\rho_a}} \tag{26}$$